Modeling Efficiency of Self-Administered Pension Funds in Zimbabwe

A Frontier-Based Optimization Analysis and Random Forest Model

H220021Y

Harare Institute of Technology

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Background to the Study

- Pension funds are major long-term institutional investors, mobilising savings and allocating capital in support of retirement income security
- Their operational effectiveness is critical for trust, fund sustainability, and timely access to benefits (Barr & Diamond, 2016)
- Frontier-based efficiency measurement has gained prominence over traditional performance indicators
- DEA, introduced by Charnes, Cooper & Rhodes (1978), is the predominant non-parametric tool for evaluating relative efficiency
- Zimbabwe's pension sector faces unique challenges:
  - Hyperinflationary crisis of 2007–2008 wiped out accumulated savings
  - Currency volatility, illiquid capital markets, constrained investment opportunities (IPEC, 2023)

Statement of the Problem

- No established optimization-based framework for assessing pension fund efficiency in Zimbabwe
- Regulatory oversight primarily emphasises compliance and solvency, not managerial efficiency
- In an economy with persistent inflation and financial instability:
  - Inefficiency leads to misallocation of long-term savings
  - Erosion of contributors' retirement benefits
  - Reduced confidence in the pension system
- **Core gap:** Absence of a rigorous, quantitative evaluation incorporating both managerial performance and scale effects using DEA

Research Objectives

1. To evaluate the **technical efficiency** of self-administered pension funds in Zimbabwe using a Data Envelopment Analysis framework
2. To **decompose** overall technical efficiency into pure technical efficiency and scale efficiency components
3. To **benchmark** individual pension fund efficiency scores against best-practice peers and rank funds relative to the efficiency frontier

Research Questions

1. Are self-administered pension funds in Zimbabwe technically efficient in utilizing their resources?
2. To what extent can overall technical inefficiency be decomposed into pure technical and scale efficiency components?
3. How do self-administered pension funds benchmark against best-practice peers on the efficiency frontier?

Significance of the Study

- **Academic:** Contributes to limited literature on pension fund efficiency in developing economies; applies frontier-based measurement framework (Kaffash et al., 2020)
- **Methodological:** Demonstrates application of non-parametric optimisation techniques (DEA) reinforcing their relevance for pension fund evaluation
- **Policy:**
  - Provides regulators with objective benchmarks beyond compliance indicators
  - Insights into resource allocation and scale optimisation for fund managers
  - Contributes to improved retirement income security

Scope and Assumptions

- **Scope:**
  - Selected registered and active **self-administered standalone** pension funds in Zimbabwe (IPEC-regulated)
  - Five-year study period: **2018–2022**
  - Limited to technical efficiency using secondary financial data
  - NSSA, insured funds, and informal sector arrangements excluded
- **Key Assumptions:**
  - Financial statements accurately represent pension fund operations
  - Selected inputs and outputs adequately capture fund activities
  - DEA assumptions regarding relative efficiency are satisfied

Research Design

- **Quantitative research design** based on Data Envelopment Analysis (DEA)
- DEA: non-parametric linear programming technique measuring relative efficiency of Decision-Making Units (DMUs)
- Each pension fund = one DMU
- **Input-output framework:**
  - **Inputs:** Total Assets (x_1), Operating Expenses (x_2), Total Equity & Debt (x_3)
  - **Outputs:** Net Investment Income (y_1), Members' Contributions (y_2)
- All ZWG figures converted to USD-equivalent at the official exchange rate on the financial statement date

Sample

- 12 registered self-administered standalone pension funds (DMU1–DMU12)
- Satisfies DEA convention: $\geq 3 \times (3 + 2) = 15$ observations at panel level
- Funds include: MIPF, POSB Staff, CBZ Bank Staff, Delta Corp, Zimplats, Air Zimbabwe, Econet Wireless, NRZ Staff, Bindura Nickel, Telecel, Innscor Africa, NMB Bank
- Source: IPEC annual regulatory reports (2018–2022)

Seven-Stage Hybrid DEA Framework

1. **Variable Selection & Data Preparation**
2. **CCR Model** — Constant Returns to Scale (CRS) DEA
3. **BCC Model** — Variable Returns to Scale (VRS) DEA
4. **Scale Efficiency Computation**

$$\mathrm{SE}_o = \frac{\theta^*_{\mathrm{CRS},o}}{\theta^*_{\mathrm{VRS},o}}$$

5. **Peer Analysis & Target Setting**
6. **Simar–Wilson Bootstrap Bias Correction** ($B = 2{,}000$)
7. **Random Forest Determinant Analysis**

DEA Models

CCR Model (CRS):

$$\max\ h_o = \frac{\sum_r u_r y_{ro}}{\sum_i v_i x_{io}} \quad \text{s.t.} \quad \frac{\sum_r u_r y_{rj}}{\sum_i v_i x_{ij}} \leq 1, \quad u_r, v_i \geq \varepsilon$$

BCC Model (VRS) adds a free scalar u_0 to capture scale effects:

$$\max\ h_o = \frac{\sum_r u_r y_{ro} + u_0}{\sum_i v_i x_{io}}$$

- $h_o = 1$: fund on the efficiency frontier
- $h_o < 1$: fund can reduce inputs proportionally
- By construction: $\theta^*_{\text{VRS}} \geq \theta^*_{\text{CRS}}$

Machine Learning Enhancements

- **Stage 6 — Simar-Wilson Bootstrap:**
  - Corrects for finite-sample upward bias in raw DEA scores
  - $B = 2{,}000$ bootstrap pseudo-samples
  - Produces bias-corrected scores $\hat{\theta}^*_{\text{BIAS}}$ with 95% confidence intervals
- **Stage 7 — Random Forest:**
  - Replaces Tobit regression for second-stage determinant analysis
  - Captures non-linear and interaction effects among macroeconomic factors
  - $T = 500$ trees; hyperparameters tuned via 5-fold cross-validation
  - Variable importance: MDI and MDA metrics
- Implementation: Python (`scikit-learn`, `scipy`)

Descriptive Statistics

Variable	Mean	Std Dev	Min	Max
Total Assets (x_1, \$m)	15.42	21.37	0.81	78.24
Operating Exp. (x_2, \$m)	1.18	1.53	0.05	5.31
Equity & Debt (x_3, \$m)	14.09	19.84	0.71	71.63
Net Inv. Income (y_1, \$m)	1.82	2.61	0.02	9.43
Members' Contr. (y_2, \$m)	2.34	3.17	0.11	11.21

- **High size heterogeneity**: Total assets range from \$0.81m to \$78.24m
- Strong inter-input correlations (> 0.90): consistent with DEA literature
- Right-skewed output distributions: small number of large funds dominate

CRS Technical Efficiency Results

- **4 funds** (33.3%) consistently achieved full CRS efficiency (score = 1.000):
  - DMU1 (MIPF), DMU3 (CBZ Bank Staff), DMU7 (Econet Wireless), DMU11 (Innscor Africa)
- Mean CRS efficiency: **0.639** — average fund could reduce inputs by ≈36.1%
- Notable efficiency decline in **2020** (avg 0.573): COVID-19 disruptions
- Recovery trend: 0.635 (2021) → 0.675 (2022)

Comparison with Greek occupational pension funds (Chrysanthopoulou et al., 2023):

- Greek average CRS: **0.730** vs Zimbabwe: **0.639**
- Reflects more severe macroeconomic constraints in Zimbabwe

VRS and Scale Efficiency Results

DMU	Fund	CRS	VRS	Scale	RTS
DMU1	MIPF	1.000	1.000	1.000	CRS
DMU2	POSB Staff	0.412	0.524	0.786	IRS
DMU3	CBZ Bank Staff	1.000	1.000	1.000	CRS
DMU4	Delta Corp	0.589	0.732	0.804	IRS
DMU5	Zimplats	0.373	0.398	0.937	DRS
DMU6	Air Zimbabwe	0.264	0.291	0.907	IRS
DMU7	Econet Wireless	1.000	1.000	1.000	CRS
DMU8	NRZ Staff	0.728	0.874	0.833	IRS
DMU9	Bindura Nickel	0.503	0.584	0.861	IRS
DMU10	Telecel	0.318	0.327	0.972	DRS
DMU11	Innscor Africa	1.000	1.000	1.000	CRS
DMU12	NMB Bank	0.484	0.566	0.855	IRS
Mean		**0.639**	**0.691**	**0.913**	

IRS = Increasing Returns to Scale; DRS = Decreasing Returns to Scale

Scale Efficiency Analysis

- Mean Scale Efficiency: **0.913**
- **6 of 8 inefficient funds** operate under Increasing Returns to Scale (IRS)
  - Operating *below* most productive scale size
  - Would benefit from consolidation and asset growth
- **2 funds** exhibit Decreasing Returns to Scale (DRS): DMU5, DMU10
  - Operating *beyond* optimal scale threshold
  - Require operational rationalisation
- Supports IPEC's regulatory objective of encouraging pension fund consolidation

Peer Analysis and Input Slacks

- **DMU7 (Econet Wireless)** is the most frequently referenced benchmark — peer for 4 inefficient funds
- DMU1 (MIPF) and DMU3 (CBZ Bank Staff) serve as peers for 3 funds each
- **Dominant slack variable: Total Equity & Debt (x_3)**
  - Primary source of over-capitalisation relative to best-practice peers
  - Consistent with Chrysanthopoulou et al. (2023) findings
- Example: DMU5 (Zimplats) must reduce total assets by \$4.52m and equity & debt by \$5.12m to reach the frontier
- Operating expenses show relatively smaller slack — cost efficiency more readily managed

Bootstrap Bias-Corrected Efficiency Scores

- Simar–Wilson bootstrap ($B = 2{,}000$) reveals average upward bias of **0.031 efficiency units**
- Bias-corrected mean CRS efficiency: **0.608** (raw: 0.639)
- 95% confidence intervals confirm genuine efficiency heterogeneity:
  - DMU7 (Econet): [0.934, 0.981]
  - DMU10 (Telecel): [0.267, 0.332] — non-overlapping
- Frontier funds (DMU1, 3, 7, 11) remain statistically distinct from inefficient funds
- Addresses the deterministic limitation of standard DEA for policy inference

Random Forest Determinant Analysis

Predictor	MDI	MDA	Effect
Exchange Rate Volatility	0.312	0.298	Negative
Annual Inflation Rate	0.287	0.271	Negative
Fund Size (log Assets)	0.198	0.214	Positive
Expense Ratio	0.112	0.108	Negative
Returns-to-Scale Class.	0.054	0.067	Mixed
Fund Age	0.014	0.013	Positive

- Model OOB $R^2 = 0.714$: explains ≈71.4% of efficiency variance
- Exchange rate volatility + inflation ≈ **56% of total variable importance**
- Macroeconomic environment is the dominant structural constraint

Research Findings

- **RQ1 — Technical Efficiency:** 8 of 12 funds (66.7%) are technically inefficient; mean bias-corrected CRS = 0.608; average fund could deliver same outputs with ≈39% fewer inputs
- **RQ2 — Efficiency Decomposition:** Mean VRS = 0.691 vs CRS = 0.639; scale effects account for a non-trivial share of total inefficiency; IRS dominates among inefficient funds
- **RQ3 — Benchmarking:** DMU7 (Econet Wireless) is the most applicable benchmark; total equity & debt is the primary excess input; confidence intervals confirm results are statistically robust

Conclusions

- Majority of Zimbabwean self-administered pension funds do **not** operate at the efficiency frontier
- Overall inefficiency decomposes into both managerial (pure technical) and structural (scale) components
- IRS among 6 of 8 inefficient funds confirms that **consolidation** is a priority policy action
- Exchange rate volatility and inflation are the **dominant macroeconomic constraints** on efficiency
- Existence of 4 frontier-efficient funds within the same environment confirms that **efficiency improvement is achievable**

Recommendations

- **For Fund Managers:**
  - Adopt peer benchmarking using DMU7 and DMU1 as best-practice references
  - Rationalise asset and capital structures to reduce equity & debt slack
- **For IPEC:**
  - Institutionalise DEA-based efficiency benchmarking in annual supervisory reporting
  - Facilitate voluntary consolidation of small IRS-classified funds
  - Enforce consistent reporting templates for DEA input–output variables
- **For National Policymakers:**
  - Prioritise macroeconomic stabilisation (exchange rate and inflation management) as a pension sector efficiency prerequisite
  - Develop multi-pillar architecture to reduce sector concentration risk

Contribution to Knowledge

- **First** DEA efficiency study of self-administered pension funds in Zimbabwe
- Introduces a **seven-stage hybrid DEA–machine learning framework** transferable to other emerging market pension studies
- Demonstrates feasibility of frontier analysis using **secondary regulatory data** in data-scarce environments
- Expands comparative evidence on pension fund efficiency in **low- and middle-income countries**

Future Research Directions

1. **Longitudinal extension** — expand panel to 10+ years
2. **Environmental DEA models** (Banker & Natarajan, 2008) — integrate macro covariates directly into the LP
3. **Network DEA** — decompose efficiency into investment management vs. administration sub-processes
4. **Cross-country SADC comparison** using meta-frontier DEA
5. **Dynamic DEA & Malmquist indices** — distinguish frontier shifts from catch-up effects

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Questions

Thank You